Kyung Ryun (Kenny) Kim

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Education

University of California, Irvine Irvine, CA
Master of Science in Computer Science *Sep 2026 – Jun 2028 (expected)*

University of California, Irvine Irvine, CA
Bachelor of Science in Software Engineering *Sep 2023 – Jun 2026*

Experience

Undergraduate Research Assistant Apr 2025 – Present
University of California, Irvine *Irvine, CA*

- Built a source-to-source C-to-Rust translation framework, designing the mutability-analysis architecture across MIR/HIR representations to infer ownership semantics from unannotated C code
- Implemented constraint-based mutability inference over Rust MIR/HIR, correctly classifying mutability for 90% of benchmark cases

Undergraduate Research Assistant June 2024 – Aug 2024
University of Osaka *Osaka, Japan*

- Conducted research on testing and evaluating LLM-based software development workflows
- Analyzed LLM-assisted development under the waterfall model to identify reliability and testing challenges
- Presented "Application of MetaGPT to Software Development" at university's academic seminar

Projects

SSA IR for Tiny language | *Rust* Jan 2026 – March 2026

- Designed and implemented an SSA-based IR with automatic phi-node insertion at control-flow join points, built on a hand-written recursive-descent parser supporting full expression grammar, functions, and nested control flow
- Implemented common subexpression elimination and copy propagation using per-block instruction tables, passing 47 test cases

Sound Ethics Capstone Project| *Python, FastAPI, TypeScript, React, Google Cloud* Sep 2025 – March 2026

- Built secure authentication and single/multi-file audio upload workflows via RESTful APIs
- Designed and deployed normalized PostgreSQL schemas (users, audio files, folders) on Google Cloud SQL

Crux-To-x86 Compiler | *Java, GitLab* Jan 2025 – March 2025

- Designed a typed intermediate representation bridging Crux's high-level semantics and x86 machine code, lowering control flow, expressions, and function calls for analysis and codegen
- Generated x86 assembly with stack-frame management and register allocation, passing a 50-test suite covering codegen and register allocation correctness

Operating System in Rust | *Rust* June 2025 – Aug 2025

- Built a bare-metal x86-64 OS kernel in Rust, implementing a 4-level page table virtual memory subsystem, CPU exception handling via an Interrupt Descriptor Table (IDT), and hardware interrupt support through the 8259 PIC
- Engineered a custom heap allocator from scratch, supporting dynamic memory allocation in a no-std environment

ZotMover (self-standing bot)| *C++, Azure* Nov 2025 – Dec 2025

- Engineered a self-balancing two-wheeled robot using an ESP32 microcontroller, L298N motor driver, and SparkFun 6DoF IMU, implementing PID control algorithms for real-time stabilization
- Developed full-stack data pipeline capturing live X/Y/Z gyroscope and accelerometer readings from the IMU, transmitting sensor data wirelessly via ESP32 to Azure for logging and analysis

Technical Skills

Programming Language: Java, Python, C/C++, Rust, JavaScript, TypeScript
Frameworks/Libraries:LLVM, React, Angular, Django, JUnit, Selenium
Database: PostgreSQL, Google Cloud SQL, AWS DynamoDB
Tools: Git, Docker, Z3